# task 5

Create:

- 1 Hub VPC
- 2 Spoke VPCs
- Connect them using VPC Peering

Ensure:

- Spoke1 can talk to Hub
- Spoke2 can talk to Hub
- Spoke1 CANNOT talk to Spoke2
- Use route tables to control traffic

## Step 1: Create VPC

- One public subnet
- One private subnet

1. Create VPC

- VPC CIDR: 10.0.0.0/16

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1. Create Subnets

Public Subnet:

- CIDR: 10.0.1.0/24
- Auto assign public IP: Enabled

Private Subnet:

- CIDR: 10.0.2.0/24
- Auto assign public IP: Disabled

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1. Create Internet Gateway

- Create IGW
- Attach to VPC

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1. Route Tables

Public Route Table:

- Add route: 0.0.0.0/0 → IGW
- Associate with public subnet

Private Route Table:

- Only local route (10.0.0.0/16 → local)
- Associate with private subnet
- Do NOT add 0.0.0.0/0

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1. Launch EC2 Instances

Bastion EC2:

- Subnet: Public subnet
- Public IP: Enabled
- Security group:
  - SSH (22) → My IP

Private EC2:

- Subnet: Private subnet
- Public IP: Disabled
- Security group:
  - SSH (22) → Bastion security group

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1. Create S3 Gateway VPC Endpoint

- VPC → Endpoints → Create endpoint
- Service: com.amazonaws.us-west-2.s3

- Type: Gateway
- Select VPC
- Select private route table
- Policy: Full access
- Create

After creation, verify in private route table:

- pl-xxxx → vpce-xxxx automatically added

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1. Ensure No NAT

- No NAT Gateway created
- Private route table must NOT contain:
  - 0.0.0.0/0 → IGW
  - 0.0.0.0/0 → NAT

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1. SSH Flow

From local:

ssh -i bastion-key.pem ubuntu@<bastion-public-ip>

From bastion:

ssh -i private-key.pem ubuntu@<private-ip>

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1. Validation

Inside private EC2:

Test internet:

curl https://google.com

Expected: fails / no response

Test S3:

Attach IAM role (AmazonS3ReadOnlyAccess) to private EC2

Then:

aws s3 ls

Expected: S3 responds (even if AccessDenied, network should work)

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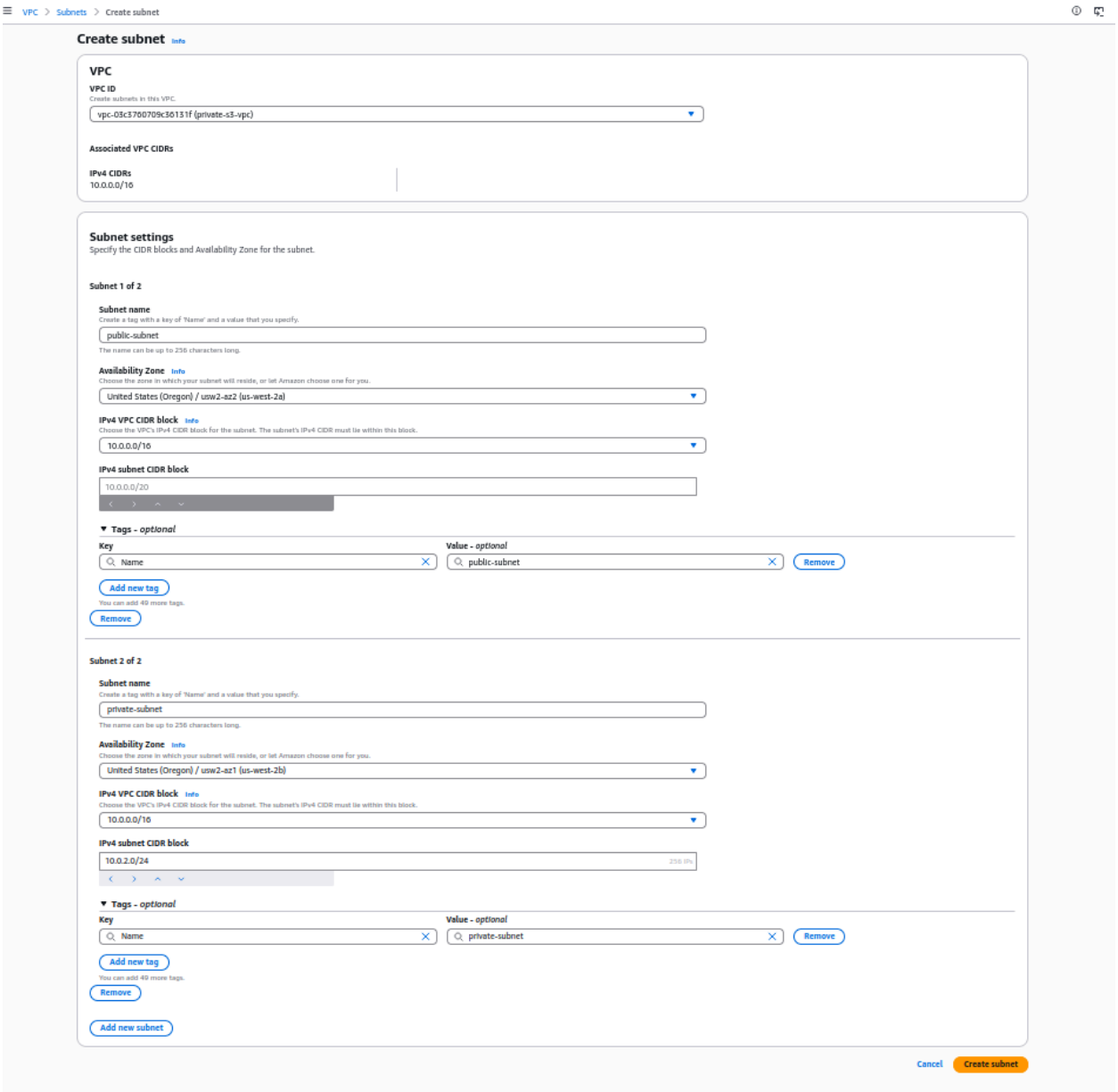
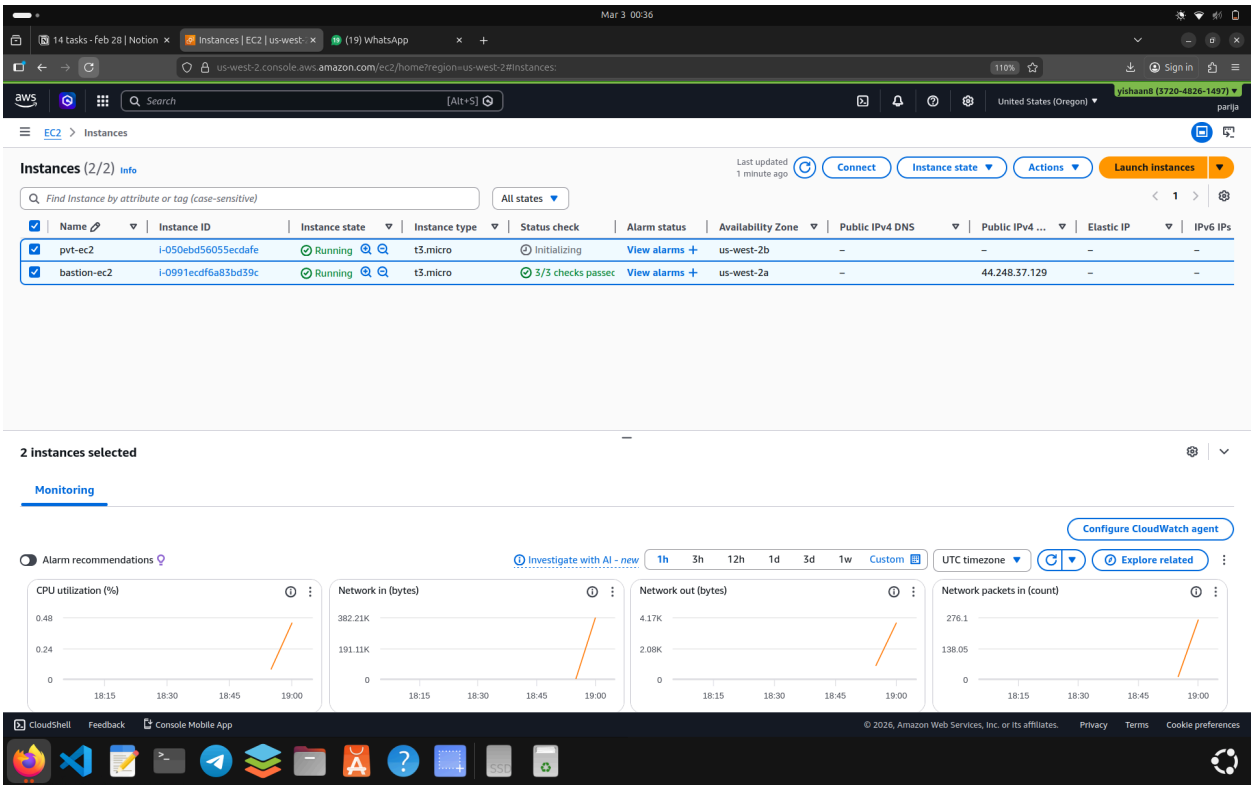
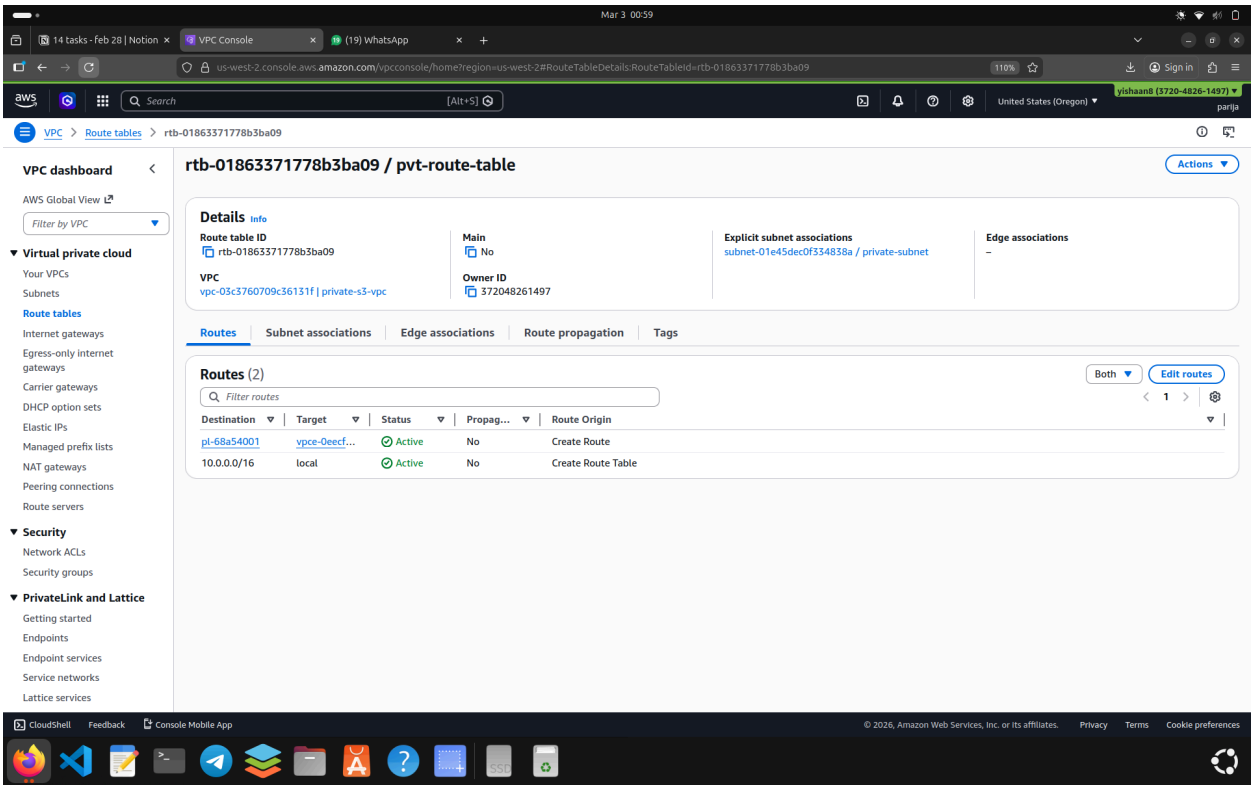
Expected Result:

- Bastion has internet
- Private EC2 has no internet
- Private EC2 can access S3 via Gateway endpoint
- No NAT used




create two ec2s - pvt and public(bastion)


Mar 3 00:36
14 tasks - feb 28 | Notion
Instances | EC2 | us-west-
(19) WhatsApp
us-west-2.console.aws.amazon.com/ec2/home?region=us-west-2#Instances:
Search
[Alt+S]
United States (Oregon)
yishaan8 (3720-4826-1497)
parija
EC2 > Instances
Instances (2/2) Info
Last updated 1 minute ago
Connect
Instance state
Actions
Launch instances
Find Instance by attribute or tag (case-sensitive)
All states
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zone
Public IPv4 DNS
Public IPv4 ...
Elastic IP
IPv6 IPs
pvt-ec2
i-050ebd56055ecdafe
Running
t3.micro
Initializing
View alarms
us-west-2b
bastion-ec2
i-0991ecdf6a83bd39c
Running
t3.micro
3/3 checks passed
View alarms
us-west-2a
44.248.37.129
2 instances selected
Monitoring
Configure CloudWatch agent
Alarm recommendations
Investigate with AI - new
1h
3h
12h
1d
3d
1w
Custom
UTC timezone
Explore related
CPU utilization (%)
Network in (bytes)
Network out (bytes)
Network packets in (count)
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Mar 3 00:59
14 tasks - feb 28 | Notion
VPC Console
(19) WhatsApp
us-west-2.console.aws.amazon.com/vpcconsole/home?region=us-west-2#RouteTableDetails:RouteTableId=rtb-01863371778b3ba09
Search
[Alt+S]
United States (Oregon)
yishaan8 (3720-4826-1497)
parija
VPC > Route tables > rtb-01863371778b3ba09
VPC dashboard
AWS Global View
Filter by VPC
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
Security groups
PrivateLink and Lattice
Getting started
Endpoints
Endpoint services
Service networks
Lattice services
rtb-01863371778b3ba09 / pvt-route-table
Actions
Details Info
Route table ID
rtb-01863371778b3ba09
Main
No
Explicit subnet associations
subnet-01e45dec0f334838a / private-subnet
Edge associations
–
VPC
vpc-03c3760709c36131f | private-s3-vpc
Owner ID
372048261497
Routes
Subnet associations
Edge associations
Route propagation
Tags
Routes (2)
Both
Edit routes
Filter routes
Destination
Target
Status
Propag...
Route Origin
pl-68a54001
vpce-0eecf...
Active
No
Create Route
10.0.0.0/16
local
Active
No
Create Route Table
CloudShell
Feedback
Console Mobile App
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Privacy
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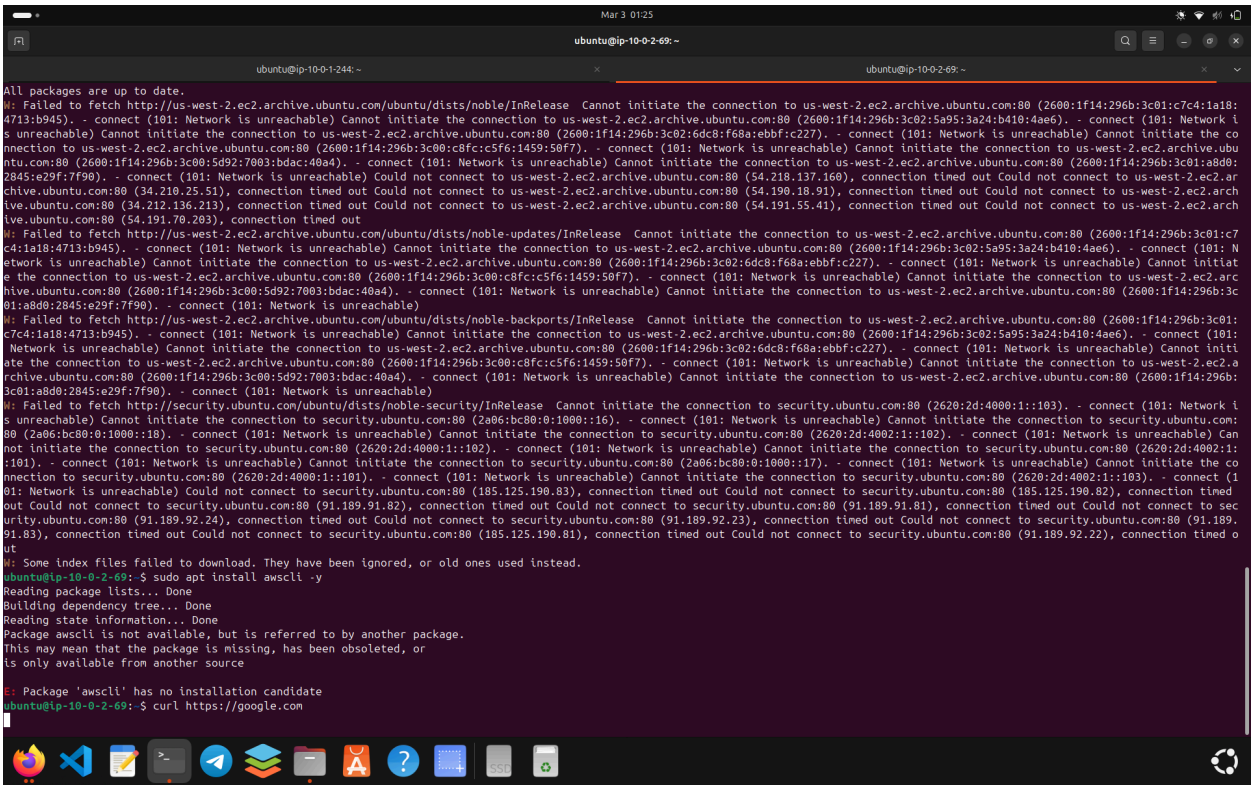

```
Mar 3  01:25
ubuntu@ip-10-0-2-69: ~
ubuntu@ip-10-0-1-244: ~        ubuntu@ip-10-0-2-69: ~
All packages are up to date.
W: Failed to fetch http://us-west-2.ec2.archive.ubuntu.com/ubuntu/dists/noble/InRelease  Cannot initiate the connection to us-west-2.ec2.archive.ubuntu.com:80 (2600:1f14:296b:3c01:c7c4:1a18:
4713:b945). - connect (101: Network is unreachable) Cannot initiate the connection to us-west-2.ec2.archive.ubuntu.com:80 (2600:1f14:296b:3c02:5a95:3a24:b410:4ae6). - connect (101: Network i
s unreachable) Cannot initiate the connection to us-west-2.ec2.archive.ubuntu.com:80 (2600:1f14:296b:3c02:6dc8:f68a:ebbf:c227). - connect (101: Network is unreachable) Cannot initiate the co
nnection to us-west-2.ec2.archive.ubuntu.com:80 (2600:1f14:296b:3c00:c8fc:c5f6:1459:50f7). - connect (101: Network is unreachable) Cannot initiate the connection to us-west-2.ec2.archive.ubu
ntu.com:80 (2600:1f14:296b:3c00:5d92:7003:bdac:40a4). - connect (101: Network is unreachable) Cannot initiate the connection to us-west-2.ec2.archive.ubuntu.com:80 (2600:1f14:296b:3c01:a8d0:
2845:e29f:7f90). - connect (101: Network is unreachable) Could not connect to us-west-2.ec2.archive.ubuntu.com:80 (54.218.137.160), connection timed out Could not connect to us-west-2.ec2.ar
chive.ubuntu.com:80 (34.210.25.51), connection timed out Could not connect to us-west-2.ec2.archive.ubuntu.com:80 (54.190.18.91), connection timed out Could not connect to us-west-2.ec2.arch
ive.ubuntu.com:80 (34.212.136.213), connection timed out Could not connect to us-west-2.ec2.archive.ubuntu.com:80 (54.191.55.41), connection timed out Could not connect to us-west-2.ec2.arch
ive.ubuntu.com:80 (54.191.70.203), connection timed out
W: Failed to fetch http://us-west-2.ec2.archive.ubuntu.com/ubuntu/dists/noble-updates/InRelease  Cannot initiate the connection to us-west-2.ec2.archive.ubuntu.com:80 (2600:1f14:296b:3c01:c7
c4:1a18:4713:b945). - connect (101: Network is unreachable) Cannot initiate the connection to us-west-2.ec2.archive.ubuntu.com:80 (2600:1f14:296b:3c02:5a95:3a24:b410:4ae6). - connect (101: N
etwork is unreachable) Cannot initiate the connection to us-west-2.ec2.archive.ubuntu.com:80 (2600:1f14:296b:3c02:6dc8:f68a:ebbf:c227). - connect (101: Network is unreachable) Cannot initiat
e the connection to us-west-2.ec2.archive.ubuntu.com:80 (2600:1f14:296b:3c00:c8fc:c5f6:1459:50f7). - connect (101: Network is unreachable) Cannot initiate the connection to us-west-2.ec2.arc
hive.ubuntu.com:80 (2600:1f14:296b:3c00:5d92:7003:bdac:40a4). - connect (101: Network is unreachable) Cannot initiate the connection to us-west-2.ec2.archive.ubuntu.com:80 (2600:1f14:296b:3c
01:a8d0:2845:e29f:7f90). - connect (101: Network is unreachable)
W: Failed to fetch http://us-west-2.ec2.archive.ubuntu.com/ubuntu/dists/noble-backports/InRelease  Cannot initiate the connection to us-west-2.ec2.archive.ubuntu.com:80 (2600:1f14:296b:3c01:
c7c4:1a18:4713:b945). - connect (101: Network is unreachable) Cannot initiate the connection to us-west-2.ec2.archive.ubuntu.com:80 (2600:1f14:296b:3c02:5a95:3a24:b410:4ae6). - connect (101:
 Network is unreachable) Cannot initiate the connection to us-west-2.ec2.archive.ubuntu.com:80 (2600:1f14:296b:3c02:6dc8:f68a:ebbf:c227). - connect (101: Network is unreachable) Cannot initi
ate the connection to us-west-2.ec2.archive.ubuntu.com:80 (2600:1f14:296b:3c00:c8fc:c5f6:1459:50f7). - connect (101: Network is unreachable) Cannot initiate the connection to us-west-2.ec2.a
rchive.ubuntu.com:80 (2600:1f14:296b:3c00:5d92:7003:bdac:40a4). - connect (101: Network is unreachable) Cannot initiate the connection to us-west-2.ec2.archive.ubuntu.com:80 (2600:1f14:296b:
3c01:a8d0:2845:e29f:7f90). - connect (101: Network is unreachable)
W: Failed to fetch http://security.ubuntu.com/ubuntu/dists/noble-security/InRelease  Cannot initiate the connection to security.ubuntu.com:80 (2620:2d:4000:1::103). - connect (101: Network i
s unreachable) Cannot initiate the connection to security.ubuntu.com:80 (2a06:bc80:0:1000::16). - connect (101: Network is unreachable) Cannot initiate the connection to security.ubuntu.com:
80 (2a06:bc80:0:1000::18). - connect (101: Network is unreachable) Cannot initiate the connection to security.ubuntu.com:80 (2620:2d:4002:1::102). - connect (101: Network is unreachable) Can
not initiate the connection to security.ubuntu.com:80 (2620:2d:4000:1::102). - connect (101: Network is unreachable) Cannot initiate the connection to security.ubuntu.com:80 (2620:2d:4002:1:
:101). - connect (101: Network is unreachable) Cannot initiate the connection to security.ubuntu.com:80 (2a06:bc80:0:1000::17). - connect (101: Network is unreachable) Cannot initiate the co
nnection to security.ubuntu.com:80 (2620:2d:4000:1::101). - connect (101: Network is unreachable) Cannot initiate the connection to security.ubuntu.com:80 (2620:2d:4002:1::103). - connect (1
01: Network is unreachable) Could not connect to security.ubuntu.com:80 (185.125.190.83), connection timed out Could not connect to security.ubuntu.com:80 (185.125.190.82), connection timed
out Could not connect to security.ubuntu.com:80 (91.189.91.82), connection timed out Could not connect to security.ubuntu.com:80 (91.189.91.81), connection timed out Could not connect to sec
urity.ubuntu.com:80 (91.189.92.24), connection timed out Could not connect to security.ubuntu.com:80 (91.189.92.23), connection timed out Could not connect to security.ubuntu.com:80 (91.189.
91.83), connection timed out Could not connect to security.ubuntu.com:80 (185.125.190.81), connection timed out Could not connect to security.ubuntu.com:80 (91.189.92.22), connection timed o
ut
W: Some index files failed to download. They have been ignored, or old ones used instead.
ubuntu@ip-10-0-2-69:~$ sudo apt install awscli -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Package awscli is not available, but is referred to by another package.
This may mean that the package is missing, has been obsoleted, or
is only available from another source

E: Package 'awscli' has no installation candidate
ubuntu@ip-10-0-2-69:~$ curl https://google.com
```